20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

Here I write normal text.

Introduction

Theory/Literature Review

Hypotheses

Analysis

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. Each dataset has to be cleaned. For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for these multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

```
# Import and inspect the raw Aargau 2009 dataset
ag_09_raw <- read_excel(
  here("data", "datasets", "aargau", "AG_09_kantonsratswahlen.xlsx"),
  sheet = "Data"
```

```
)

glimpse(ag_09_raw)
```

```
Rows: 241
Columns: 21
$ bfsNr        <dbl> 19, 1901, 1902, 1903, 1904, 1905, 1906, 1907, 1908, 1909,…
$ year         <dbl> 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 2009, 200…
$ month        <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, …
$ day          <dbl> 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, 8, …
$ locationName <chr> "Kanton Aargau", "Bezirk Aarau", "Bezirk Baden", "Bezirk …
$ locationType <chr> "CANTON", "DISTRICT", "DISTRICT", "DISTRICT", "DISTRICT",…
$ TOTAL        <dbl> 1830693, 240252, 705906, 205249, 111613, 63448, 44191, 12…
$ SVP          <dbl> 556964, 72796, 189366, 66189, 36228, 23440, 15559, 45138,…
$ SP           <dbl> 291085, 47330, 106510, 27604, 17504, 8360, 5717, 21596, 5…
$ CVP          <dbl> 275911, 21319, 131935, 39901, 10580, 2554, 11438, 8054, 1…
$ FDP          <dbl> 264828, 43245, 99220, 25582, 21661, 10744, 4886, 17379, 4…
$ GRUENE       <dbl> 164610, 23528, 64744, 16035, 12601, 5952, 3607, 7866, 306…
$ EVP          <chr> "83096", "14365", "28138", "3522", "6931", "5581", "941",…
$ SD           <chr> "21239", "2463", "6757", "1950", "...", "2719", "788", "1…
$ EDU          <chr> "30609", "7227", "6544", "1603", "1574", "2649", "368", "…
$ BDP          <chr> "71489", "...", "41512", "11629", "...", "...", "...", "7…
$ UNS          <chr> "1196", "1196", "...", "...", "...", "...", "...", "...",…
$ GLP          <chr> "68113", "6783", "30882", "10055", "4534", "1449", "887",…
$ NPAK         <chr> "76", "...", "...", "...", "...", "...", "...", "...", "…
$ L_PLUS       <chr> "1179", "...", "...", "1179", "...", "...", "...", "...",…
$ FAAG         <chr> "298", "...", "298", "...", "...", "...", "...", "...", "…
```

```
print(ag_09_raw, n = 20)
```

```
# A tibble: 241 × 21
   bfsNr  year month   day locationName locationType  TOTAL    SVP     SP    CVP
   <dbl> <dbl> <dbl> <dbl> <chr>        <chr>         <dbl>  <dbl>  <dbl>  <dbl>
 1    19  2009     3     8 Kanton Aarg… CANTON       1.83e6 556964 291085 275911
 2  1901  2009     3     8 Bezirk Aarau DISTRICT     2.40e5  72796  47330  21319
 3  1902  2009     3     8 Bezirk Baden DISTRICT     7.06e5 189366 106510 131935
 4  1903  2009     3     8 Bezirk Brem… DISTRICT     2.05e5  66189  27604  39901
 5  1904  2009     3     8 Bezirk Brugg DISTRICT     1.12e5  36228  17504  10580
 6  1905  2009     3     8 Bezirk Kulm  DISTRICT     6.34e4  23440   8360   2554
 7  1906  2009     3     8 Bezirk Lauf… DISTRICT     4.42e4  15559   5717  11438
 8  1907  2009     3     8 Bezirk Lenz… DISTRICT     1.25e5  45138  21596   8054
```

```
 9  1908  2009     3     8 Bezirk Muri  DISTRICT     4.27e4  13893   5225  12752
10  1909  2009     3     8 Bezirk Rhei… DISTRICT     6.70e4  20511  12840  10660
11  1910  2009     3     8 Bezirk Zofi… DISTRICT     1.75e5  54312  32663  11971
12  1911  2009     3     8 Bezirk Zurz… DISTRICT     5.01e4  19532   5736  14747
13  4001  2009     3     8 Aarau        TOWNSHIP     6.81e4  13936  16727   6550
14  4271  2009     3     8 Aarburg      TOWNSHIP     1.30e4   3189   3484   1419
15  4221  2009     3     8 Abtwil       TOWNSHIP     1.20e3    432     53    521
16  4191  2009     3     8 Ammerswil    TOWNSHIP     1.70e3    716    355     60
17  4222  2009     3     8 Aristau      TOWNSHIP     1.52e3    384    214    462
18  4061  2009     3     8 Arni (AG)    TOWNSHIP     3.85e3   1194    340    705
19  4272  2009     3     8 Attelwil     TOWNSHIP     1.34e3    667     63     77
20  4091  2009     3     8 Auenstein    TOWNSHIP     3.82e3    876    532    175
# ℹ 221 more rows
# ℹ 11 more variables: FDP <dbl>, GRUENE <dbl>, EVP <chr>, SD <chr>, EDU <chr>,
#   BDP <chr>, UNS <chr>, GLP <chr>, NPAK <chr>, L_PLUS <chr>, FAAG <chr>
```

```
names(ag_09_raw)
```

```
 [1] "bfsNr"        "year"         "month"        "day"          "locationName"
 [6] "locationType" "TOTAL"        "SVP"          "SP"           "CVP"
[11] "FDP"          "GRUENE"       "EVP"          "SD"           "EDU"
[16] "BDP"          "UNS"          "GLP"          "NPAK"         "L_PLUS"
[21] "FAAG"
```

```
# The district seat magnitudes and the official 2009 party seat totals
# are taken from the official Aargau election documentation and are
# hard-coded here as fixed inputs for the counterfactual exercise.

# THEREFORE LOOK IF POSSIBLE TO FIND DATASET WITH THE SEAT MAGNITUDES AND THE
OFFICIAL 2009 PARTY SEAT TOTALS
ag_09_district_seats <- c(
  "Bezirk Aarau" = 16L,
  "Bezirk Baden" = 30L,
  "Bezirk Bremgarten" = 16L,
  "Bezirk Brugg" = 11L,
  "Bezirk Kulm" = 9L,
  "Bezirk Laufenburg" = 7L,
  "Bezirk Lenzburg" = 12L,
  "Bezirk Muri" = 7L,
  "Bezirk Rheinfelden" = 10L,
  "Bezirk Zofingen" = 15L,
```

```
  "Bezirk Zurzach" = 7L
)

ag_09_parties <- c(
  "svp", "sp", "cvp", "fdp", "gruene", "evp", "sd",
  "edu", "bdp", "uns", "glp", "npak", "l_plus", "faag"
)

# Keep the district results and convert "..." (party did not stand) to zero
ag_09_districts <- ag_09_raw |>
  clean_names() |>
  filter(location_type == "DISTRICT") |>
  mutate(
    district_seats = unname(ag_09_district_seats[location_name]),
    across(
      c(total, all_of(ag_09_parties)),
      ~ replace_na(as.numeric(na_if(as.character(.x), "...")), 0)
    )
  )

# Aargau's 2005 Hagenbach-Bischoff allocation, applied separately
# within each district. The main counterfactual assumes no list alliances.
allocate_hb <- function(votes, n_seats) {
  distribution_number <- floor(sum(votes) / (n_seats + 1)) + 1
  seats <- floor(votes / distribution_number)

  while (sum(seats) < n_seats) {
    quotients <- votes / (seats + 1)
    winner <- which.max(quotients)
    seats[winner] <- seats[winner] + 1L
  }

  as.integer(seats)
}

ag_09_hb_by_district <- ag_09_districts |>
  select(location_name, district_seats, all_of(ag_09_parties)) |>
  pivot_longer(
    cols = all_of(ag_09_parties),
    names_to = "parties",
    values_to = "party_votes"
  ) |>
  group_by(location_name) |>
```

```
  mutate(
    seats_hb = allocate_hb(party_votes, first(district_seats)),
    voter_number = floor(
      party_votes / first(district_seats) + 0.5
    )
  ) |>
  ungroup()

# Official party seat totals under Double Pukelsheim in 2009
ag_09_dp_seats <- tribble(
  ~parties, ~seats_dp,
  "SVP",          45L,
  "SP",           22L,
  "CVP",          21L,
  "FDP",          20L,
  "GRUENE",       13L,
  "EVP",           6L,
  "SD",            2L,
  "EDU",           2L,
  "BDP",           4L,
  "UNS",           0L,
  "GLP",           5L,
  "NPAK",          0L,
  "L_PLUS",        0L,
  "FAAG",          0L
)

# Final election-party dataset
ag_09_clean <- ag_09_hb_by_district |>
  group_by(parties) |>
  summarise(
    voter_number = sum(voter_number),
    seats_hb = sum(seats_hb),
    .groups = "drop"
  ) |>
  mutate(
    parties = str_to_upper(parties),
    vote_share = 100 * voter_number / sum(voter_number),
    small_party = as.integer(vote_share < 10)
  ) |>
  left_join(ag_09_dp_seats, by = "parties") |>
  transmute(
    election_id = "AG_09",
```

```
    parties,
    vote_share = round(vote_share, 4),
    small_party,
    seats_hb = as.integer(seats_hb),
    seats_dp = as.integer(seats_dp)
  ) |>
  arrange(desc(vote_share))

# Validation and inspection
stopifnot(
  identical(
    names(ag_09_clean),
    c("election_id", "parties", "vote_share", "small_party",
      "seats_hb", "seats_dp")
  ),
  nrow(ag_09_clean) == length(ag_09_parties),
  sum(ag_09_clean$seats_hb) == 140L,
  sum(ag_09_clean$seats_dp) == 140L,
  abs(sum(ag_09_clean$vote_share) - 100) < 0.01
)

glimpse(ag_09_clean)
```

```
Rows: 14
Columns: 6
$ election_id <chr> "AG_09", "AG_09", "AG_09", "AG_09", "AG_09", "AG_09", "AG_…
$ parties     <chr> "SVP", "SP", "CVP", "FDP", "GRUENE", "EVP", "GLP", "BDP", …
$ vote_share  <dbl> 31.9206, 15.7320, 15.0231, 14.3039, 8.8875, 4.4681, 3.4599…
$ small_party <int> 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
$ seats_hb    <int> 54, 24, 23, 20, 10, 4, 2, 3, 0, 0, 0, 0, 0, 0
$ seats_dp    <int> 45, 22, 21, 20, 13, 6, 5, 4, 2, 2, 0, 0, 0, 0
```

```
print(ag_09_clean, n = Inf)
```

```
# A tibble: 14 × 6
   election_id parties vote_share small_party seats_hb seats_dp
   <chr>       <chr>        <dbl>       <int>    <int>    <int>
 1 AG_09       SVP           31.9           0       54       45
 2 AG_09       SP            15.7           0       24       22
 3 AG_09       CVP           15.0           0       23       21
 4 AG_09       FDP           14.3           0       20       20
```

```
 5 AG_09       GRUENE      8.89             1       10       13
 6 AG_09       EVP         4.47             1        4        6
 7 AG_09       GLP         3.46             1        2        5
 8 AG_09       BDP         3.06             1        3        4
 9 AG_09       EDU         1.76             1        0        2
10 AG_09       SD          1.24             1        0        2
11 AG_09       UNS         0.0641           1        0        0
12 AG_09       L_PLUS      0.0633           1        0        0
13 AG_09       NPAK        0.0094           1        0        0
14 AG_09       FAAG        0.0086           1        0        0
```

For the counterfactual, the 2009 district-level party votes and the 2009 district seat magnitudes are retained. The Hagenbach–Bischoff allocation follows Sections 12–13 of the Aargau Grossratswahlgesetz used for the 2005 election: the distribution number is the next integer above the valid party votes divided by the number of seats plus one; initial seats are the integer quotients; and remaining seats are assigned sequentially using the largest quotient of party votes divided by seats already received plus one (Kanton Aargau, 2004, pp. 6–7). Although Section 8 allowed list alliances in 2005, the main counterfactual treats every party independently because list alliances were abolished for the 2009 Double Pukelsheim election. The district seat magnitudes are the official assignments for the 2009 election (Staatskanzlei des Kantons Aargau, 2008), while `seats_dp` records the officially reported result (Statistisches Amt des Kantons Aargau, 2009). Vote shares use the cantonal *Wählerzahl*: each district's party votes are divided by its seat magnitude and rounded to the nearest integer before cantonal aggregation (Grossratswahlgesetz, Section 14).

Results

Interpretation of the Results

Discussion

Conclusion

Bibliography

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Statistisches Amt des Kantons Aargau. (2009, April 21). *Detailergebnisse der Grossratswahlen vom 8. März 2009*. https://www.ag.ch/de/medien/medienmitteilungen?mm=detailergebnisse-der-grossratswahlen-vom-8-maerz-2009-742f7d21-d4b0-454f-abd7-25471ef7f566_de

Staatskanzlei des Kantons Aargau. (2005, 7. März). Bezirksprotokolle über die Erneuerungswahl des Grossen Rats für die Amtsperiode 2005/2009: Wahltag: 27. Februar 2005 (Amtsblattbeilage 2005-1). https://doczz.net/doc/5954970/amtsblattbeilage_vom_07.03.2005

Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance